Target dependent interval coverage in donor nested gene module contrasts

A simulation study with an existing data illustration

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Abstract

Background: Donor-aware analysis addresses dependence between cells, but interval interpretation also depends on module membership and weighting. We examined whether intervals covered complete-module, observed-member or population cell-weighted effects in a fixed continuous gene-module model.

Methods: A prospectively frozen main catalog contained 192 scenarios with 5000 replicates each, varying donor correlation, donor number, membership coverage, noise, background overlap and informative cell numbers. Paired cell-level and equal-donor Welch analyses used identical inputs. Primary evaluation concerned complete-donor-target coverage; observed-donor and population-cell targets, matching-null errors, power and four-module false discovery rate were reported separately. An existing eight-donor primary biliary cholangitis (PBC) derivative provided a separate descriptive illustration.

Results: The simulation produced 960000 instances with no module failures. Across correlated A-block null labels, cell-level Type I error was 55.98%-80.20%, compared with 2.66%-5.42% for donor means; cell error was 4.56%-5.78% at the zero-donor-correlation boundary. In a prespecified partial-membership scenario with 12 donors per group and sparse signal in a five-member module, donor coverage was 94.90% for the observed target but 88.54% for the complete target. The secondary paired difference was 6.36 percentage points (95% Monte Carlo interval, 5.52-7.20). Under informative cell numbers, population-target interpretation did not restore cell-level interval calibration. In the PBC derivatives, cell-count weighting reversed two of four contrast directions, while all four inherited maxT-adjusted P-values remained 1.0.

Conclusions: In these finite continuous mechanisms, donor aggregation addressed dependence but did not guarantee complete-module coverage under changing membership. Target choice and interval calibration require separate evaluation. Sparse signal was tested only in one five-member module; the PBC illustration supplied neither biological truth nor independent validation. The results do not establish a new scoring method or a disease mechanism.

Keywords: donor-nested data; gene modules; estimands; confidence interval coverage; simulation; informative cluster size.

Introduction

Single-cell studies often obtain many observations from relatively few individuals. Cells within an individual share biological and sampling influences, so increasing the number of cells does not necessarily increase the number of independent experimental units. Zimmerman and colleagues examined this dependence and its implications for differential-expression testing.[1] Squair and colleagues likewise highlighted false discoveries and the value of retaining biological-replicate structure.[2] These studies already establish the importance of avoiding pseudoreplication; demonstrating the same problem in a module contrast would not, by itself, constitute a new statistical contribution. Subsequent benchmarking also shows why method comparisons must specify performance criteria and simulated inputs rather than reducing performance to a single error measure.[3]

Moving from individual genes to module scores introduces another interpretive question: which collection of genes defines the effect being estimated? A complete gene set and its observed subset can represent different contrasts, even when both are summarized within donors. Scoring methods also operate on different scales. For example, UCell uses within-cell ranks rather than a fixed-scale arithmetic expression mean.[4] Consequently, robustness of one scoring algorithm cannot be assumed to imply coverage of a complete-module effect for another. Here, we use a deliberately simple arithmetic module mean to define complete and observed targets explicitly, not to replace established signature-scoring tools or model an end-to-end transcriptomic pipeline.

Weighting adds a related distinction. Giving each donor equal weight and giving each cell equal weight answer different questions when donors contribute different numbers of cells. When those numbers are related to donor states, the distinction is not merely one of precision. Informative cluster size has an established methodological literature, including multilevel within-cluster resampling approaches.[5] In a finite donor sample, a cell-weighted ratio estimator can also have an expectation different from its population ratio target. Choosing that population target does not by itself account for within-donor dependence in an interval calculation. Target selection, finite-sample estimator behavior and interval calibration therefore need to be examined separately.

We investigated these questions using a frozen continuous gene-level model with known truth. Two simple Welch analyses were applied to the same inputs across 192 scenarios, and each interval was evaluated against complete-donor, observed-donor and population-cell targets. The primary question was complete-donor-target coverage; matching-target diagnostics were retained to distinguish target disagreement from errors against a method's own null. We also retained all background and boundary scenarios, rather than selecting only conditions unfavorable to a cell-level analysis. Finally, previously observed PBC donor derivatives illustrated weighting and member sensitivity without providing a new disease test or independent biological validation. The intended contribution is a bounded, quantitative account of target-dependent interval behavior, not

a universal method ranking, a new inferential correction or a mechanistic claim about PBC.

Methods

Study design and estimands

We evaluated whether confidence intervals for gene-module contrasts retained their intended interpretation when the analysis unit, observed members or cell-number weights changed. The study comprised a known-truth simulation and a previously observed, donor-level primary biliary cholangitis (PBC) illustration. Simulation reporting followed the aims, data-generating mechanisms, estimands, methods and performance measures structure.[6] The design was developed after exposure to earlier score-level pilot results and the PBC derivatives. Main scenarios, seeds, targets, failure rules and primary display coordinates were then frozen before the main simulation was executed. The main simulation was not a blinded, wholly outcome-naive development process.

For a module with complete membership M and observed membership O, the complete donor target, theta_D_full, was the mean effective group coefficient across M. The observed donor target, theta_D_observed, was the corresponding mean across O. The population cell target, theta_C_observed, additionally incorporated the group contrast in cell-number-weighted donor states. The recorded target drift was theta_D_observed minus theta_D_full. A cell-weighted population target was distinguished from the finite-sample expectation of the ratio estimator. Both methods' intervals were assessed against all three targets; matching-target diagnostics used theta_D_observed for the donor method and theta_C_observed for the cell-level method.

Gene level generation membership and scenario factors

We generated continuous float64 expression for 49 anonymous virtual gene positions partitioned into four disjoint modules of size 5, 11, 16 and 17. For donor d, cell c and gene g, expression was

$$X_{dcg} = \beta_g G_d + \sqrt{\rho} U_d + \sqrt{\gamma} V_{dc} + \sqrt{1-\rho-\gamma} E_{dcg}$$

where G_d indicated the case group. The unit-variance donor factor U was shared across a donor's cells and all modules; the cell factor V was shared across that cell's genes; and gene residuals E were independent. We fixed gamma=0.2 and varied rho between 0 and 0.4. Thus rho was the donor variance component, not the intraclass correlation of every resulting module score. Gaussian and unit-variance Laplace noise were examined. Laplace scale was 1/sqrt(2). No count sampling, library-size normalization, sample-wise standardization or rank-based signature scoring was applied to the simulated expression.

Direct signal was confined to M1, the five-member module. Uniform signal assigned beta=delta to every M1 member. Sparse signal assigned beta=delta×5/2 to its first two members and zero to the other three, retaining a complete-module contrast delta. Delta was 0 or 0.4. Full coverage retained all members; partial coverage sampled ceil(0.6K) members independently of the expression errors, giving 3, 7, 10 and 11 observed positions. Each replicate had one random membership mask per module shared by all donors. Signal

locations were fixed by gene position, not selected from the generated data.

The 192 catalog scenarios comprised three blocks. Block A crossed donor number per group (4 or 12), rho, fixed cell-count layout (20 per donor or a repeating 5/20/80 sequence), signal structure, coverage, delta and noise family, yielding 128 scenarios. Block B contained 32 scenarios with no direct core signal, rho=0.4 and the repeating layout; it varied donor number, coverage, noise, background overlap and background effect. Background overlap involved the first floor(K/4) members of each module, with coefficient eta=0 or 0.4. Background coefficients entered the effective expression target rather than being reclassified as statistical null effects.

Block C contained 32 scenarios with rho=0.4 and uniform direct signal. Donor states were sqrt(3/2) times -1, 0 or 1 with equal probability. Cell counts were 5, 20 or 80. In the independent rule, counts were independent of donor states; in the informative rule, counts increased with state in cases and decreased with state in controls. We varied donor number, coverage, delta, count rule and Gaussian/Laplace cell and gene noise. Under the informative rule, the population cell-target shift was 1.1065666703. Exact finite-distribution calculations fixed the cell estimator's finite-n expected shifts at 0.8475456225 and 1.0436783188 for 4 and 12 donors per group, respectively. These were auxiliary diagnostics, not replacement primary estimands.

Paired methods intervals and multiplicity

Each cell's score was the arithmetic mean of its observed module members. The cell-level method applied a two-sided Welch comparison to all cell scores. The donor method first averaged cell scores within each donor, then applied Welch's test to equally weighted donor means. Both methods used the same expression realization and membership masks. Cell observations were therefore not counted as additional biological replicates in the donor analysis.

Welch standard errors used each group's sample variance divided by its observation count, with Satterthwaite degrees of freedom. Intervals were the estimate plus or minus the 0.975 Student-t quantile times the standard error. The cell method used cell counts in these expressions; the donor method used donor counts. Two-sided raw $P<0.05$ defined rejection. Benjamini-Hochberg adjustment was applied separately within each method and replicate across the four modules at q=0.05.[7] False discovery proportion was the number of rejected matching-target nulls divided by max(1, total rejections); empirical false discovery rate (FDR) was its replicate mean. The study evaluated the adjustment under the simulated dependence structure rather than assuming universal error control.

Performance measures and Monte Carlo uncertainty

The primary performance measure was empirical 95% interval coverage of theta_D_full. We also reported coverage and bias against the observed donor and population cell targets, interval length, root mean squared error, and paired method differences. Matching-target Type I error was conditional on the matching target being null, using absolute truth tolerance 10^{-12}; power was conditional on a non-null matching target. Random sparse masks could therefore change the applicable Type I and power denominators within a scenario. Rejection fields repeated across target rows were not treated as independent observations.

Each scenario contained 5000 independent replicates. For binary measures, Monte Carlo standard error (MCSE) was sqrt[p(1-p)/n], accompanied by a Wilson 95% Monte Carlo interval. FDR uncertainty used the sample variance of replicate false discovery proportions; paired differences used the sample variance of same-input differences. Missing eligible denominators were recorded as not applicable rather than zero. A failed module retained its failure reason and unavailable P/interval fields; a failed four-module family did not receive an artificial zero false discovery proportion.

After the main results were available, we calculated secondary, within-method coverage differences between targets on the unchanged 16 primary Figure3 coordinates. These diagnostics used 80000 existing instances and paired binary coverage indicators, with MCSE and mean±1.96 MCSE. They added neither simulations nor a new primary endpoint. Descriptive extrema across the full design were labelled as post-main findings, not substituted for the preselected primary coordinates. Scenario labels that reduced to the same effective generating mechanism were retained but not pooled as independent biological mechanisms.

Execution and numerical checking

The master seed was 202610137400. Stable scenario/replicate identifiers and separate streams for masks, donor states, counts and cell/gene noise defined random inputs; method identity did not enter seed generation. Main input/code identities were frozen before execution, separately from engineering seeds. Python 3.12.14 and NumPy 2.3.5 were used. Eight fixed worker partitions changed scheduling only. Records were retained in immutable 250-replicate gzip blocks, together with their identities and summaries.

Production t probabilities used an incomplete-beta continued fraction and quantiles used bracketing/bisection. Local verification read every main record, independently reconstructed target labels, binary coverage/error fields, BH decisions and array-based summaries, and regenerated replicates 0, 2499 and 4999 in every scenario. Input regeneration used the same frozen generator; scoring and scalar Welch calculations used a separate calculation path. Gauss-Legendre density integration provided a different algorithm for selected P-value and interval-probability checks. This was numerical verification by a single local execution process, not external scientific validation. Editable SVG and 300-dpi PNG author-review figures were generated using Matplotlib 3.11.2 from verified source rows.

Previously observed PBC derived data illustration

The illustration used existing local donor-level log-counts-per-million (logCPM) derivatives from the processed 5-prime archive OMIX001122-99-02.[8] The upstream resource combined single-cell and spatial data, but only the previously derived single-cell donor tables were used here. The fixed primary subset comprised four controls and four PBC donors; the additional PBC donor PBC2_5 was not promoted into that subset. We did not newly reconstruct the full count matrix or re-annotate cells.

The existing reconstruction standardized each gene across the eight primary donors using the sample standard deviation and averaged the fixed eligible members of four ciliary modules: anterograde intraflagellar transport (IFT), retrograde IFT, basal body/distal appendage and transition zone. The reconstruction was required to agree with the existing scores within 10^{-6}. Equal-donor contrasts were PBC minus control. A descriptive alternative weighted the same donor scores by each donor's consensus cell count within group; it was not a raw-count pseudobulk reanalysis or a new cell-level significance test.

The existing studentized permutation analysis enumerated all 70 four-versus-four allocations and used the maximum absolute statistic across four modules for maxT family-wise adjustment. These inherited results were preserved. Leave-one-member-out analysis kept the original gene-standardization scale fixed and recomputed module contrasts after each deletion. The 49 correlated deletion operations were summarized by minima, maxima and direction changes; their ranges were not confidence intervals. Figure aliases mapped explicitly to the unchanged original donor keys. Clinical covariate adjustment, unknown biological truth and independent external validation were outside this illustration.

Post main analytical explanation and virtual data reproduction

After completion of the main simulation, we derived the distribution of the observed target for the frozen sparse M1 mask and decomposed equal-donor contrast variance under the A/B fixed-count mechanisms. All ten three-of-five masks were enumerated using exact fractions and compared with the finite-population variance expression. Conditional noise variance was checked independently with donor-block cell-score covariance matrices. Correspondence to existing mean squared error summaries was descriptive, not a joint threshold requiring every diagnostic to lie inside its individual 95% Monte Carlo interval. The A/B independence formula was not applied to informative C counts. C population ratios and finite-sample ratio expectations were checked by factorial-weighted compositions and a separate integer convolution. These explanations did not introduce a corrected interval or replace a primary target.

A virtual-only reproduction packet was tested offline in a separate directory using the pinned local runtime. The original generator and numerical code were reused unchanged; runtime roots and package-level source guards were injected explicitly. Fixed IDs 0, 2499 and 4999 in each scenario were replayed against existing canonical records. Complete scenario summaries were joined to the exported tables with their frozen parameter annotations. Synthetic figures were rebuilt with unchanged plotting instructions after excluding patient-dependent components. This was fixed-ID numerical reproduction within

the same operating system, not new scientific replication, an independent generator or a rerun of all main raw records. Full derivations and boundaries are provided in the analytical supplement.

Results

Complete main execution retained all scenarios and targets

All 192 scenarios completed 5000 replicates, yielding 960000 unique instances and 1920000 paired method records. The 3840 retained blocks passed full readback. No module failed. The complete outputs comprised 4608 scenario-method-module-target rows, 384 method-family rows and 2304 paired-method rows. These records retained every scenario, including conservative and non-generalizing results (Figure1; Supplementary result tables).

Verification regenerated 576 inputs and compared 192592 summary fields. The largest absolute summary discrepancy was 1.87×10^{-14}. The largest discrepancies for selected P-values and interval probabilities between numerical algorithms were 8.56×10^{-9} and 3.31×10^{-9}, respectively. These checks supported record and numerical consistency within the tested implementation; they were not biological validation.

Donor correlation changed null calibration whereas the zero correlation boundary remained visible

Across A-block M1 null labels with rho=0.4, cell-level Type I error ranged from 55.98% to 80.20%, compared with 2.66%-5.42% for donor means. At rho=0, the respective ranges were 4.56%-5.78% and 2.46%-5.36%. Donor means were sometimes conservative rather than exactly calibrated at 5% (Figure2; Supplementary Figures S1 and S4). These are ranges over fixed design labels, not confidence intervals for a population of studies.

In preselected scenario A048, with four donors per group, correlated donor states and a repeating cell-count layout, cell-level coverage of the full donor target was 19.80% (95% Monte Carlo interval, 18.72%-20.93%). Its raw Type I error was 80.20% and empirical four-module FDR was 80.78%. Donor coverage was 96.18%, with Type I error 3.82% and FDR 2.82%. In the corresponding rho=0 scenario A016, cell-level coverage was 94.40% and Type I error 5.60%. The observed loss of calibration therefore depended on the generating structure, rather than on the cell-level label alone.

Observed member coverage did not guarantee full module coverage

Within the prespecified sparse-signal coordinates, donor intervals covered the observed target more often than the full target under partial membership. In A126, with 12 donors per group and three of five M1 members observed, coverage was 94.90% for theta_D_observed (95% Monte Carlo interval, 94.25%-95.48%) and 88.54% for theta_D_full (87.63%-89.39%; Figure3a). The secondary paired coverage difference was 6.36 percentage points (MCSE, 0.426 points; normal 95% Monte Carlo interval, 5.52-7.20 points). Mean bias against the full target was 0.00276 (MCSE, 0.00476), while matching-target FDR was 3.16%. Near-zero average bias did not imply correct full-target interval coverage.

The difference was smaller at four donors per group: A062 had full/observed coverage of 94.76%/96.04%, a paired difference of 1.28 points (95% Monte Carlo interval, 0.80-1.76). In A126, masks generated 524 matching-target null instances and 4476 non-null instances; raw Type I error and power were 5.73% and 36.97% in those distinct subsets. Cell-level power was higher, but its full/observed-target coverage was only 20.48%/23.42% (Figure3b,d). Power alone did not distinguish target alignment or calibration.

Partial masks produced observed M1 effects of 0, 1/3 or 2/3 around a complete effect of 0.4; full masks had zero target drift (Figure3c). Complete fixed-design sensitivity showed substantial heterogeneity. The post-main descriptive maximum gap occurred in A078, where donor full/observed coverage was 50.10%/95.00%; its full-member counterpart A074 had coverage 95.18%. A046, by contrast, had partial full/observed coverage 95.00%/96.20%. These full-design findings were retained in Supplementary Figure S5 and did not replace the primary rho=0.4 display coordinates.

Exact target variation explained a component of complete target uncertainty

For sparse M1 with coefficients (1,1,0,0,0), the ten equally probable three-member masks produced observed targets 0, 1/3 and 2/3 with probabilities 1/10, 6/10 and 3/10. Their mean was exactly 0.4 and variance exactly 0.04. In A126, the conditional equal-donor contrast noise variance was 103/1440 (0.0715277778), while complete-target total variance was 803/7200 (0.1115277778). The added component came from membership-target variation shared across donors. This explains how an unbiased point estimator across masks can have uncertainty not represented by within-arm dispersion; it does not derive the empirical Welch coverage percentages or establish a new interval correction.

The exact A/B variance calculations agreed with direct covariance-matrix calculations across 640 scenario-module rows. Two exact C calculations reproduced finite-n shifts 0.8475456225 and 1.0436783188 at n=4 and n=12, while preserving the population shift 1.1065666703. These were post-main deterministic explanations of the existing generator. The separate offline packet matched all 576 existing replay IDs and complete summary tables, and reconstructed seven synthetic figures with 14 PNG/SVG byte matches. Patient-derived inputs were not required for this reproduction.

Informative cell numbers separated population and finite sample targets

In informative, full-membership C scenarios with no direct group effect, the population cell target differed from the full donor target by 1.10657. At four donors per group, the mean cell-level estimate in C002 was 0.85133, close to the finite-n expected shift 0.84755 rather than the population target. Its bias against the population cell target was -0.25524 (MCSE, 0.00673). At 12 donors per group, C018 had a mean estimate of 1.04784 and population-target bias -0.05873 (MCSE, 0.00265), consistent with a smaller finite-n discrepancy (Figure3f).

Interval calibration remained a separate issue. Cell-level coverage was 6.66%/0.06% against the full donor target in C002/C018, and 41.92%/43.36% against the population cell target. Donor full-target coverage was 95.14%/94.62% (Figure3e). Across all 32 C scenarios, donor full-target coverage ranged from 94.60% to 95.80%, whereas cell matching-population-target coverage ranged from 23.04% to 45.20% (Supplementary Figure S3).

Cell matching-target FDR was zero in C002 and C018 because all four population cell targets were non-null. The low full-donor-target coverage in these scenarios was therefore not a Type I error against the cell method's own non-null target. Changing the interpretive target also did not restore its interval coverage.

Background expression effects were not direct core effects

The B block had zero direct core coefficients, but overlapping background effects created nonzero effective expression targets. In full-membership B012, the four targets were 0.0800, 0.0727, 0.1000 and 0.0941. Both methods consequently had matching-target FDR zero, while M1 cell/donor coverage was 20.16%/95.40%. This distinction preserved expression truth without relabelling background overlap as a direct core effect. All 32 background scenarios and four modules were retained (Supplementary Figure S2 and complete result tables).

Existing PBC derivatives illustrated weighting and membership sensitivity without a disease positive result

The eight primary donors contributed 1041 consensus cells: 852 from controls and 189 from PBC donors. Individual contributions ranged from 24 to 441 cells. Equal-donor weighting assigned 25% within each group, whereas the largest cell-count weight was 51.76% among controls (Figure4a). This illustration used eight donors, not 1041 independent participants.

Cell-count weighting reversed two of four module contrasts. Anterograde IFT changed from -0.0861 to 0.0253, and basal body/distal appendage from -0.0543 to 0.3156. Retrograde IFT and transition-zone contrasts retained their directions. All four inherited maxT-adjusted P-values were 1.0; no new significance test was attached to the weighted contrasts (Figure4b).

Fixed-scale member deletion produced two direction reversals within the five-member basal body/distal appendage module and one within the 16-member transition-zone

module. Their deletion ranges were -0.1728 to 0.1083 and -0.00686 to 0.1537, respectively (Figure4c). The 17-member anterograde and 11-member retrograde modules had no deletion-induced reversals. These correlated perturbations illustrated numerical sensitivity of existing derived scores, not causal roles of omitted genes or independent validation of the simulated mechanisms.

Discussion

Donor aware analysis and target alignment answer different questions

The central result is that interval coverage depended on the target assigned to the interval, not only on whether analysis respected the donor unit. Donor means addressed the dependence represented in the model, while random partial membership could leave intervals aligned with observed members rather than the complete module. Informative cell numbers created a second separation between donor and cell-population effects, with a further finite-sample ratio discrepancy. These are distinct sources of interpretation failure. The simulation quantified them on paired inputs, and the existing-data illustration showed that weighting and member choices can change derived contrasts without producing disease-positive evidence.

The correlated null scenarios agree with the established pseudoreplication concern described by Zimmerman and Squair.[1][2] They should not be presented as its discovery. The zero-donor-correlation boundary is equally important: cell-level calibration was substantially different there, and donor intervals were sometimes conservative. Thus, the evidence supports a dependence-specific explanation rather than a rule that every cell-level analysis is invalid. Our donor method is a Welch comparison of continuous module means, not a normalized count-pseudobulk model or a mixed-effects model. The two-method experiment cannot settle the wider rankings debated in differential-expression benchmarks, where performance criteria and modeling choices differ.[3]

Unbiased averages can coexist with poor complete target coverage

The sparse-member result is not explained by a large average donor-estimator bias. A random subset can average to the complete target over repeated masks while targeting a different effect in an individual replicate. In the prespecified A126 coordinate, observed-target coverage remained near nominal while complete-target coverage was lower. The paired secondary diagnostic confirmed a within-interval difference rather than a comparison of unrelated simulations. This explains why reporting a small aggregate bias alongside an observed-member confidence interval is insufficient to establish coverage of a full module.

This interpretation remains specific to the membership mechanism studied. One mask was shared across donors and sampled independently of expression errors. Direct sparse signal was present only in M1, with two affected positions among five; the other module sizes did not receive equivalent signal experiments. The study therefore does not establish a general law relating module size to signal robustness. Nor does it simulate gene-specific detectability, condition-dependent missingness, dropout or an uncertain biological gene set. The much larger post-main gap in A078 is informative about heterogeneity within the

frozen design, but it does not replace the prespecified coordinate used in the abstract and main display. No new interval correction or optimal member-selection procedure was evaluated.

The post-main variance decomposition makes this distinction explicit. In the frozen sparse module, membership-target variance was 0.04 even though the target averaged to the complete effect. Expression uncertainty decreased with the donor/count configuration, whereas the mask was shared across all donors and its target variation did not average away. Accordingly, the within-arm uncertainty used by a Welch comparison need not include that shared target displacement. The result explains a source of the observed discrepancy without proving exact Welch coverage, proposing a calibrated correction or extending the sparse experiment beyond M1.

Weighting finite samples and error rates must retain their own targets

The informative-count block connects to the established cluster-size problem, rather than introducing it.[5] Within the frozen discrete mechanism, equal-donor and equal-cell analyses genuinely target different populations. A cell-weighted effect need not be scientifically inappropriate if that is the question of interest. However, the empirical cell estimator at a small donor sample tracked its finite-sample ratio expectation more closely than the population ratio target, and its naive cell-level intervals remained poorly calibrated even against the population-cell target. Relabelling an interval is therefore not a substitute for constructing an interval that accounts for the sampling structure.

The false discovery results provide a parallel caution. FDR is defined with respect to statistical nulls, not absence of a named mechanism.[7] A family of non-null cell-population targets can have zero false discoveries even while its intervals poorly cover the full donor target. Similarly, background overlap produced nonzero expression targets despite zero direct core coefficients. Those detections were not statistical false positives under the effective expression target, but they did not demonstrate a direct core mechanism. Retaining the explicit null definition prevents both favorable zero-FDR results and unfavorable target discrepancies from being misclassified.

The PBC illustration supports sensitivity analysis not mechanistic validation

The originating PBC resource investigated cholangiocyte biology using several experimental modalities.[9] Our illustration does not re-evaluate those biological conclusions. It uses an already selected donor subset and existing logCPM derivatives, with no new count reconstruction, annotation audit or independent cohort. Equal-donor and cell-count-weighted means of the same scores reversed two contrast directions; fixed-scale member deletion also altered some directions. These operations make numerical choices visible, but unknown biological truth prevents assessment of which contrast is correct. All four inherited maxT-adjusted P-values were 1.0, so the illustration cannot support a disease-positive ciliary result.

There is also a scale difference between the simulation and illustration. Simulated genes had a fixed theoretical scale, whereas the PBC scores used gene-wise standardization across the same eight donors. Deletion kept that original scale fixed and did not provide

independent replicates or sampling confidence intervals. Weighting reused donor scores rather than reconstructing a cell-level or count-pseudobulk analysis. Accordingly, concordance of a sensitivity pattern with the simulated examples is illustrative, not evidence that the simulation's missing-member or informative-count mechanisms generated the biological data.

Scope reproducibility and practical interpretation

The main experiment is limited to continuous expression with two noise families, two simple analyses and a small catalog of donor/count configurations. It omits count sampling, sequencing-depth normalization, nonlinear or rank-based scoring, overlapping biological modules and broad cell-state composition changes. In particular, no UCell benchmark was run, and its rank-score properties cannot be inferred from these arithmetic-mean results.[4] Retaining all scenarios, explicit conditional denominators and Monte Carlo uncertainty makes the finite evidence inspectable; it does not make it representative of all transcriptomic studies. The design was informed by earlier pilot outcomes and the existing PBC derivatives, although main inputs and primary display coordinates were frozen before main execution. Numerical readback and alternate calculation paths support implementation consistency, not external scientific validation. These distinctions accord with transparent simulation reporting.[6]

A practical implication is to state the analysis unit, gene membership, weighting rule and inferential target before interpreting a module contrast. Reporting matching-target calibration alongside complete-target coverage helps expose questions that a single P-value or FDR summary cannot answer. For these models, donor aggregation is useful but insufficient to guarantee a complete-module interpretation under random sparse membership. The post-main derivations and virtual-only reproduction packet now provide a model-specific explanation and a tested route to the existing numerical results. They do not extend the evaluated scoring methods, biological validity or operating-system coverage.

Conclusion

In the tested continuous gene-module models, respecting donor dependence did not by itself guarantee interval coverage for the complete module when observed membership changed. Cell weighting further separated donor, population-cell and finite-sample estimator targets. The useful inference is to specify and evaluate these targets separately, not to equate detection with mechanism. Sparse signal was evaluated only in a five-member module, and the existing PBC derivatives remained a negative, descriptive illustration rather than biological validation. The study supports a bounded computational account of target-dependent coverage, not a general scoring method or disease-mechanism claim.

Declarations and availability

Author contributions: Zhi Chen and Teng Qi approved the final manuscript and accompanying materials. Specific CRediT roles remain to be confirmed from the actual work performed.

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Ethics and consent: The authors declare that ethics approval was not required for this study. The computational experiments used synthetic data. The descriptive PBC illustration reused existing local donor-level derivatives from the cited resource; no new participants or specimens were collected. No approval identifier, consent waiver or new data-access application is claimed.

Data and code availability: The synthetic reproduction packet and this manuscript release are provided through https://github.com/1209433622cz-maker/ctro-claimgate. The packet contains the frozen configuration, full aggregate tables, all 192 scenario summaries and 576 existing replay fixtures. Its tested offline entry point reproduces the fixed-ID subset, table joins, seven synthetic figures and analytical explanation using the pinned runtime. It does not contain or rerun all 960000 raw records, demonstrate cross-operating-system portability or implement an independent generator. The full raw simulation archive remains local and is not part of the present deposit. No Zenodo DOI has yet been assigned to this release.

The PBC source is OMIX001122, processed archive OMIX001122-99-02. The public synthetic packet does not contain the original patient matrices or local donor-level input derivatives. The PBC figure is an author-created descriptive illustration, not an independent validation dataset.

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Figure legends

Main figures

Figure1. Design and estimands of the continuous gene-module simulation and separate derived-data illustration. (a) The generator contained 49 virtual gene positions in four disjoint modules; direct signal was confined to M1 (K=5). (b) Complete donor, observed donor and population cell targets were defined separately. (c) The frozen catalog contained 128 A, 32 B and 32 C scenarios, each with 5000 replicates. (d) Cell-level and equal-donor Welch analyses used paired inputs. (e) The previously observed PBC illustration used existing donor derivatives, not a new count-matrix analysis or independent external validation. Boxes describe computational components, not biological causal pathways.

Abbreviations: BH, Benjamini-Hochberg; CI, confidence interval; FDR, false discovery rate; K, full module membership size; PBC, primary biliary cholangitis.

Figure2. Null calibration on eight preselected coordinates. Panels display (a) full-donor-target interval coverage, (b) matching-target raw Type I error and (c) empirical FDR after four-module BH adjustment. Orange squares represent cell-level Welch and blue circles equal-donor Welch. All panels use M1, uniform zero signal, Gaussian noise and the repeating 5/20/80 cell-count layout. Labels specify scenario ID, donors per group, rho and full/partial coverage. Each point is based on 5000 instances. Error bars in a/b are Wilson 95% Monte Carlo intervals; c uses mean±1.96 MCSE, not a binary-proportion interval. Dashed references are 95% coverage and 5% error/FDR. These exact coordinates were frozen before main execution; the full design is retained in the supplements.

Abbreviations: BH, Benjamini-Hochberg; FDR, false discovery rate; MCSE, Monte Carlo standard error; n, donors per group; r, donor variance component rho. "All" denotes full membership.

Figure3. Member coverage and informative cell numbers distinguish estimands. Panels a-d use the eight frozen sparse-M1, delta=0.4, rho=0.4, repeating-layout coordinates: A058/059 have n=4 and full membership; A062/063 n=4 and partial membership; A122/123 n=12 and full membership; A126/127 n=12 and partial membership. The first/second ID in each pair uses Gaussian/unit-variance Laplace noise. (a,b) The same donor/cell intervals are evaluated against complete and observed donor targets; filled circles indicate complete targets and open triangles observed targets. (c) Stacked bars show empirical frequencies of observed-minus-complete target drift, including the zero-drift full-membership reference. (d) Raw power is conditional on a non-null matching target; denominators differ with sparse masks and are recorded in source rows. Panels e/f use the eight frozen full-membership, delta=0 C coordinates: C000/001 are n=4 independent counts; C002/003 n=4 informative counts; C016/017 n=12 independent counts; C018/019 n=12 informative counts, again Gaussian/Laplace pairs. (e) Cell intervals are evaluated against the complete donor and population cell targets, and donor intervals against the complete donor target. (f) Cell estimator means are compared with analytically defined population and finite-n expectation targets. All rates use 5000 instances except conditional power; coverage/power bars are Wilson 95% Monte Carlo intervals, and estimator-mean bars are ±1.96 MCSE. Some bars are smaller than their symbols. No population target was replaced with its finite-n expectation as a primary endpoint.

Abbreviations: MCSE, Monte Carlo standard error; M1, five-member virtual module; n, donors per group. Orange denotes the cell method and blue the donor method; black crosses denote population targets and purple plus signs finite-n expectations.

Figure4. Weight and member sensitivity in previously observed PBC donor derivatives. (a) Within-group cell-count weights of eight unchanged primary donors; numbers above bars are consensus cell counts. Control/PBC A-D are display aliases, not new donor IDs; the source register maps them to CTR1_5, CTR2_5, CTR3_5, CTR4_1_5 and PBC1_5, PBC3_5, PBC4_5, PBC5_1_5. The dashed line marks equal-donor weight, 25%. (b) PBC-minus-control contrasts from equal-donor and cell-count-weighted means of the same existing scores. Weighting is descriptive, not a raw-count reanalysis or new significance test. All four inherited maxT P-values are 1.0. (c) Points show complete-member contrasts and lines the minimum-to-maximum range after deleting each member on a fixed gene-standardization scale. K and the number of deletion-induced sign reversals are listed. The 49 deletions are correlated sensitivity operations; lines are not confidence intervals. Gene scores used the eight-donor sample-z scale derived from existing logCPM, unlike the fixed theoretical scale of the simulation.

Abbreviations: IFT, intraflagellar transport; K, module member count; logCPM, log-counts per million; maxT, maximum absolute studentized permutation statistic; PBC, primary biliary cholangitis. "Flips" denotes deletion-induced direction reversals. This illustration provides no new biological truth or independent external validation.

Figure1

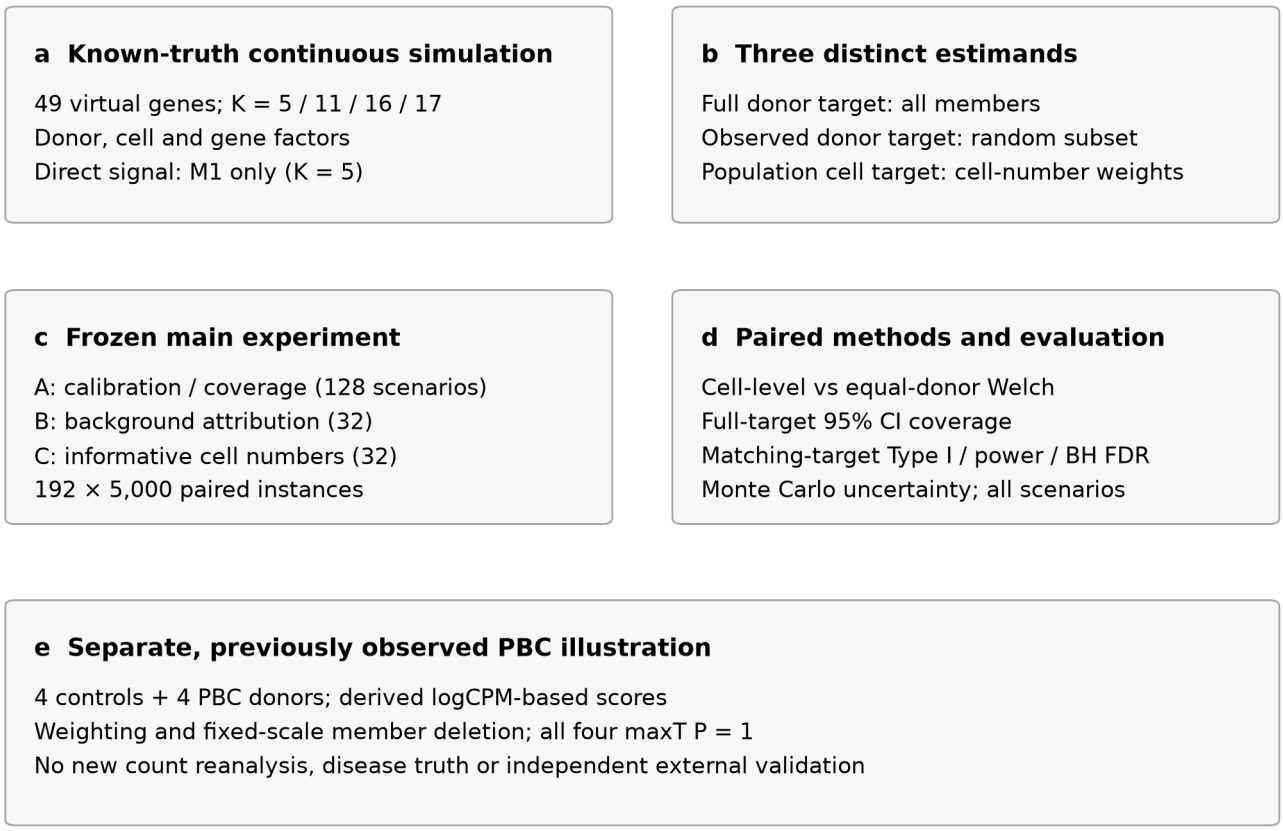

a Known-truth continuous simulation
49 virtual genes; K = 5 / 11 / 16 / 17
Donor, cell and gene factors
Direct signal: M1 only (K = 5)
b Three distinct estimands
Full donor target: all members
Observed donor target: random subset
Population cell target: cell-number weights
c Frozen main experiment
A: calibration / coverage (128 scenarios)
B: background attribution (32)
C: informative cell numbers (32)
192 × 5,000 paired instances
d Paired methods and evaluation
Cell-level vs equal-donor Welch
Full-target 95% CI coverage
Matching-target Type I / power / BH FDR
Monte Carlo uncertainty; all scenarios
e Separate, previously observed PBC illustration
4 controls + 4 PBC donors; derived logCPM-based scores
Weighting and fixed-scale member deletion; all four maxT P = 1
No new count reanalysis, disease truth or independent external validation

Figure2

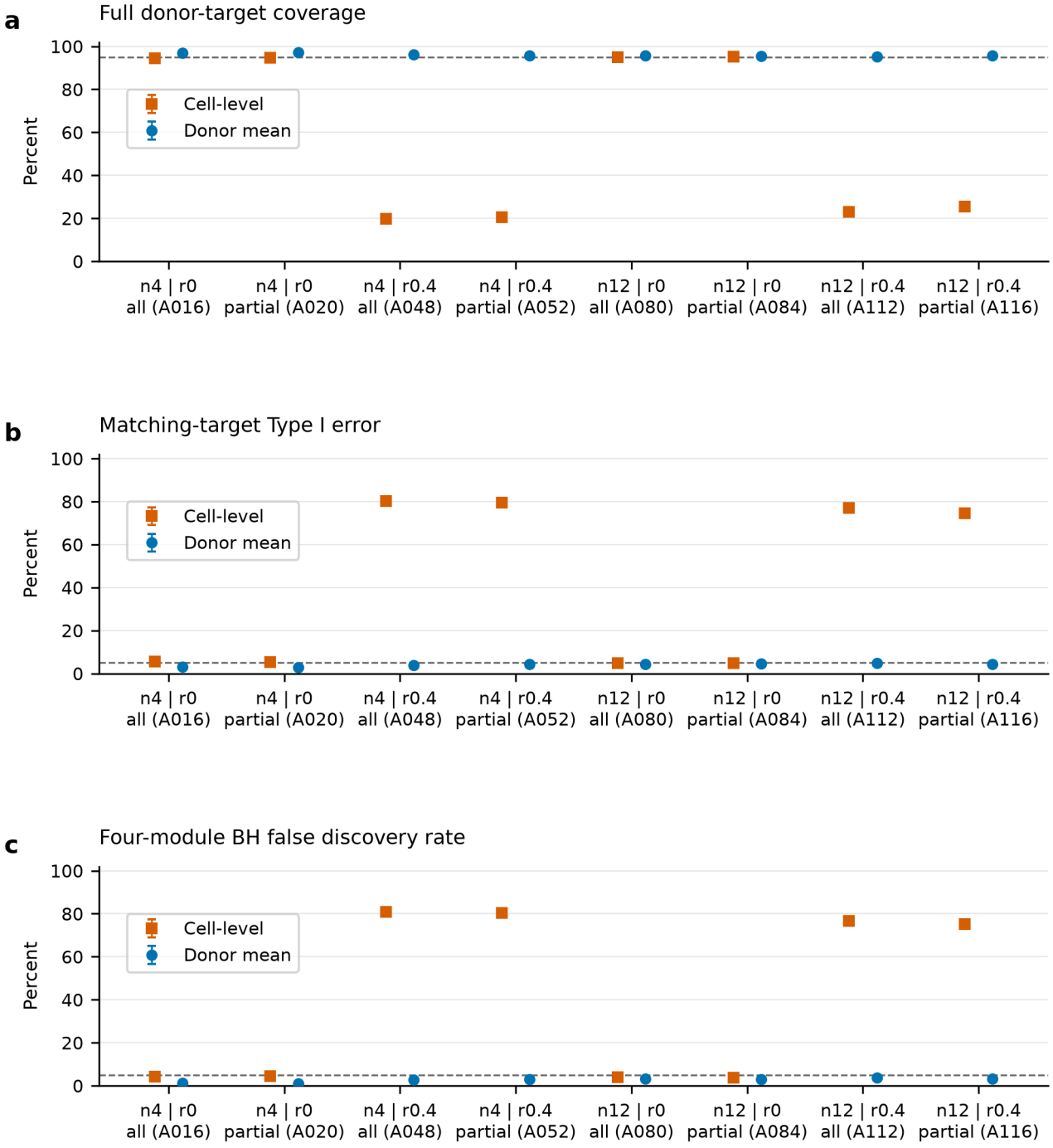

a
Full donor-target coverage
Percent
Cell-level
Donor mean
b
Matching-target Type I error
Percent
Cell-level
Donor mean
c
Four-module BH false discovery rate
Percent
Cell-level
Donor mean
n4 | r0 all (A016)
n4 | r0 partial (A020)
n4 | r0.4 all (A048)
n4 | r0.4 partial (A052)
n12 | r0 all (A080)
n12 | r0 partial (A084)
n12 | r0.4 all (A112)
n12 | r0.4 partial (A116)

Figure3

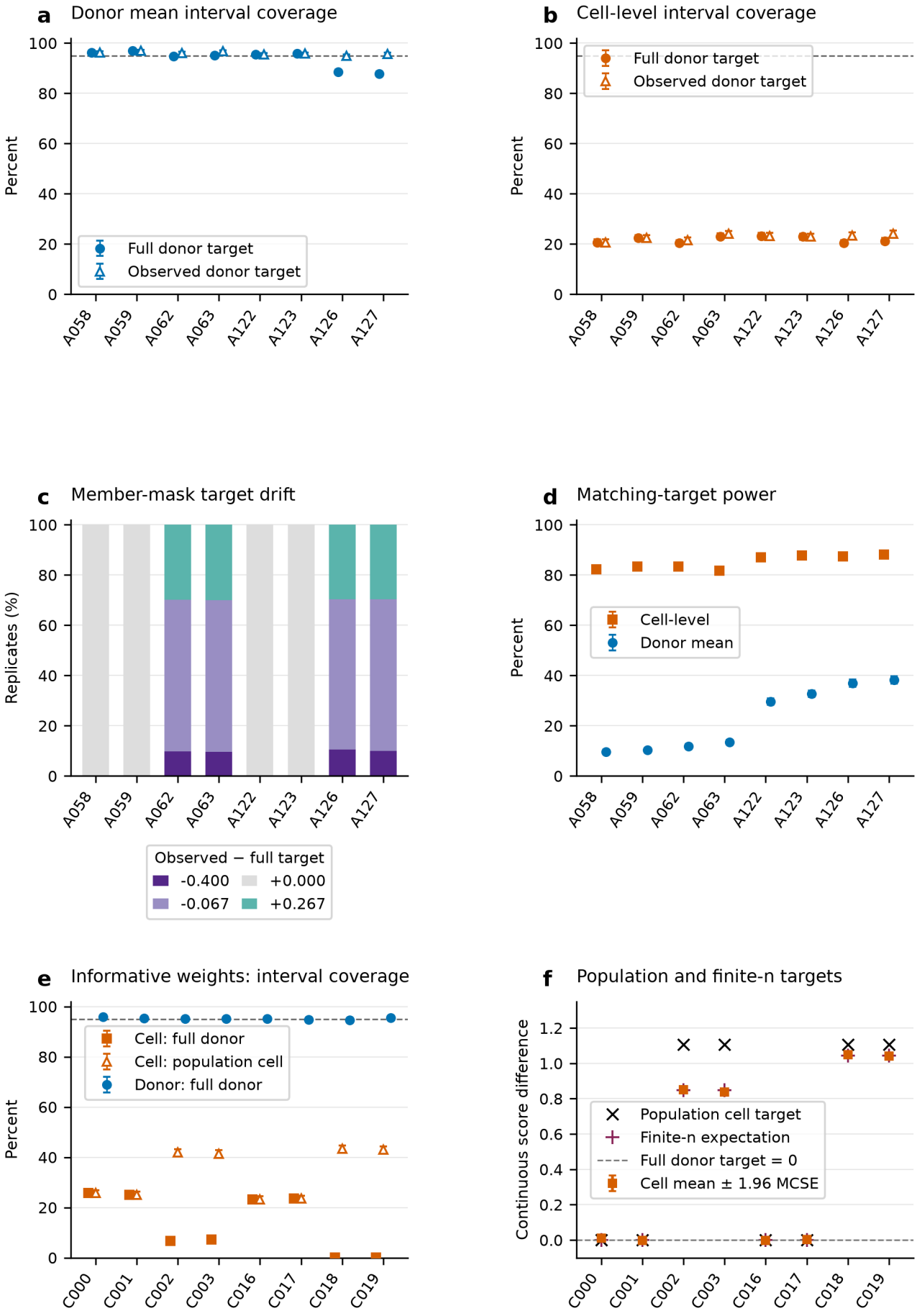
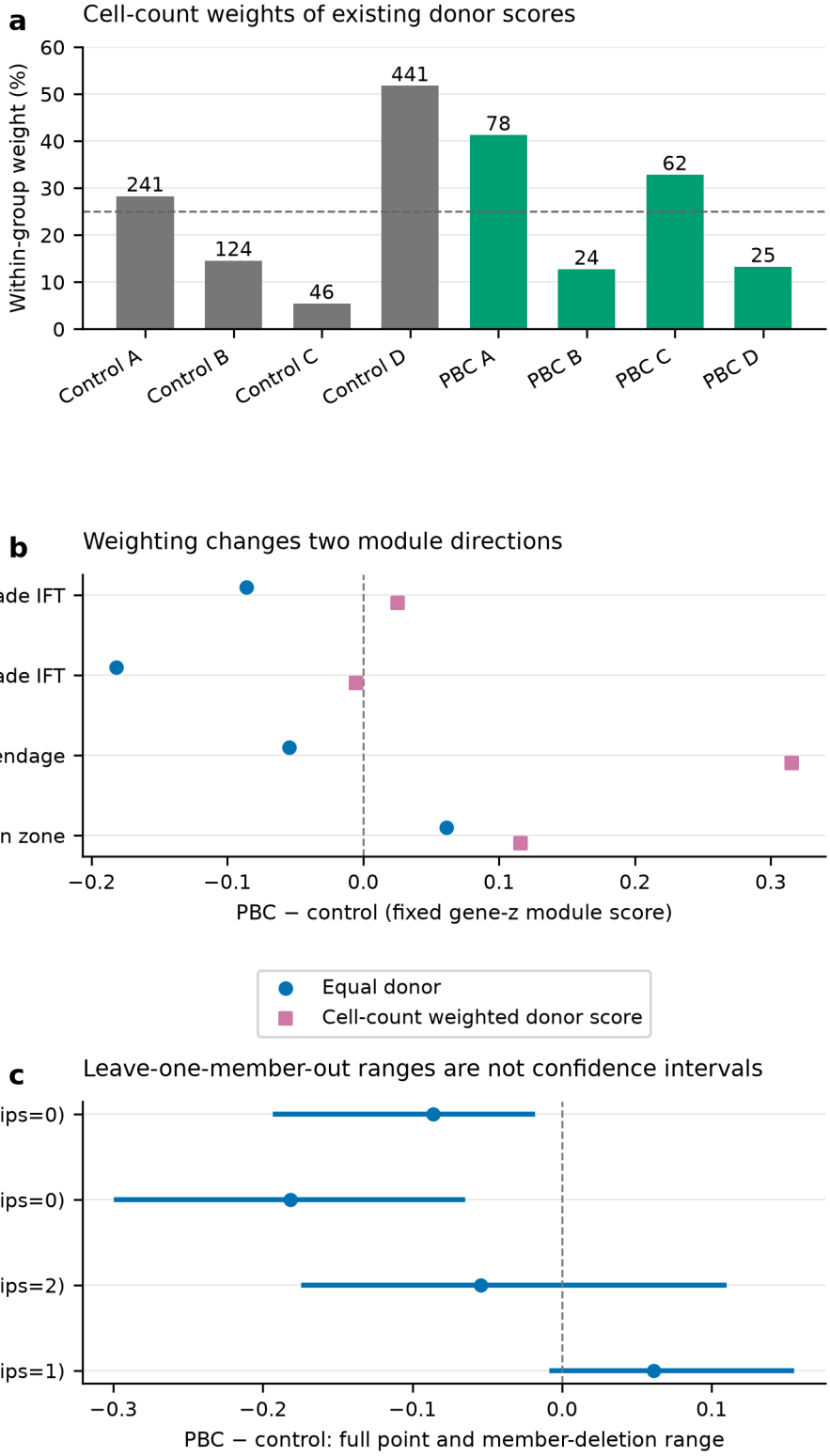

a
Donor mean interval coverage
Percent
Full donor target
Observed donor target
b
Cell-level interval coverage
Full donor target
Observed donor target
Percent
c
Member-mask target drift
Replicates (%)
Observed − full target
-0.400
+0.000
-0.067
+0.267
d
Matching-target power
Percent
Cell-level
Donor mean
e
Informative weights: interval coverage
Percent
Cell: full donor
Cell: population cell
Donor: full donor
f
Population and finite-n targets
Continuous score difference
Population cell target
Finite-n expectation
Full donor target = 0
Cell mean ± 1.96 MCSE
A058
A059
A062
A063
A122
A123
A126
A127
C000
C001
C002
C003
C016
C017
C018
C019

Figure4


a
Cell-count weights of existing donor scores
Within-group weight (%)
241
124
46
441
78
24
62
25
Control A
Control B
Control C
Control D
PBC A
PBC B
PBC C
PBC D
b
Weighting changes two module directions
Anterograde IFT
Retrograde IFT
Basal body / distal appendage
Transition zone
PBC − control (fixed gene-z module score)
Equal donor
Cell-count weighted donor score
c
Leave-one-member-out ranges are not confidence intervals
Anterograde IFT (K=17, flips=0)
Retrograde IFT (K=11, flips=0)
Basal body / distal appendage (K=5, flips=2)
Transition zone (K=16, flips=1)
PBC − control: full point and member-deletion range